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Service Network System for an Operating Theatre

Abstract

A service network system **1**, for providing services to an operating theatre facility **3**, comprising, an air handling unit **19**, an interface pre-assembled module **13**, a receiver pre-assembled module **11**, an operating theatre pre-assembled module **9**, ventilation supply ductwork **50,55,77** and ventilation return ductwork **52,57,79**. The ventilation supply ductwork **50, 55,77** and ventilation return ductwork **52, 57,79** pass from the air handling unit **9** then through the interface pre-assembled module **13**, then through the receiver pre-assembled module **11** and then into the operating theatre pre-assembled module **9.**

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a service network system for a traditionally built operating theatre or for a modular operating theatre and in particular to a service network system for an operating theatre provided with an ultra-clean ventilation system. The utility requirements for these areas are complex and require a streamlined solution to reduce the construction time.

BACKGROUND

[0002] The construction industry is known to have a chronically low level of productivity that has changed very little in the past seventy years. It is thought that the low level of productivity is a consequence of the generally accepted need for bespoke buildings to be developed for particular applications combined with 'tried-and-tested' construction methods.

[0003] Bespoke buildings can take several years to design, construct and validate which is slow and by implication costly. Furthermore, building designs are also often subject to changes as a project progresses and, in some circumstances, buildings must be further modified after completion to comply with a change in requirements, expansion or cost constraints.

[0004] The problem is particularly acute for research and medical facilities which often need to be constructed rapidly and which must be flexible to accommodate changes to their end use. For example, laboratory facilities such as vaccine development facilities, advanced therapy medicinal product facilities and cell and gene therapy facilities are often planned and initiated very quickly, with a need for them to be constructed and commissioned in several months rather than several years and to be adaptable during and after construction.

[0005] In addition, once a building has been constructed, it is typically time consuming and costly to install service networks of conveying components, such as ventilation systems, piping networks and electrical networks, in the building.

[0006] It can be desirable to locate plant equipment, such as HVAC systems, electrical generators and the like, on the roof of a facility rather than inside a facility. This can save space, provide easier access to plant equipment for installation, maintenance and inspection purposes, and make it easier to reconfigure the facility by adding, removing or replacing plant equipment as needed.

[0007] However, locating plant equipment on the roof of a facility can have several disadvantages. It can increase project cost, complexity and duration by requiring a bespoke design and installation to connect plant equipment with services on the roof and to the service network inside the facility. It can also make the facility less easily reconfigurable.

[0008] There is a need to provide a modular building system that addresses some or all of the above-described disadvantages. In relation to the present invention, the modular building system approach is used to provide a modular operating theatre.

SUMMARY OF THE INVENTION

[0009] In accordance with a first aspect of the invention there is provided a service network system, for providing services to an operating theatre facility, comprising, an air handling unit, an interface pre-assembled module, a receiver pre-assembled module, an operating theatre pre-assembled module, ventilation supply ductwork and ventilation return ductwork, wherein the ventilation supply ductwork and ventilation return ductwork pass from the air handling unit then through the interface pre-assembled module, then through the receiver pre-assembled module and then into the operating theatre pre-assembled module. This arrangement is advantageous because it avoids having to build services on site, which is inevitably time consuming and costly. In addition, providing a receiver pre-assembled structure that is separate to an operating theatre pre-assembled structure facilitates road transportation of the service network system from its manufacturing location to the site where it will be installed.

[0010] Preferably, the operating theatre facility is located within a building and wherein the air handling unit is located outside of the building, the interface pre-assembled module is located in an external surface of the building such that it has a first part that is located outside of the building and a second part that is located inside of the building, the receiver pre-assembled module is located within the building and the operating theatre pre-assembled module is located within the building, wherein the ventilation supply ductwork and ventilation return ductwork pass through the interface pre-assembled module. The provision of a pre-assembled interface module is advantageous because it facilitates passing of all of the services through the external fabric of the building through a single aperture.

[0011] Preferably, the air handling unit is located on the roof of the building and vertically above the operating theatre pre-assembled module, wherein the operating theatre pre-assembled module and the receiver pre-assembled module are located adjacent to each other and the ventilation supply ductwork and the ventilation return ductwork run laterally between the operating theatre pre-assembled module and the receiver pre-assembled module. This facilitates the routing of all of the ventilation ductwork through the pre-assembled interface module, thus avoiding having to make multiple apertures through the roof of the building.

[0012] Preferably, the interface pre-assembled module is a lid pre-assembled module.

[0013] Preferably, the operating theatre pre-assembled module is provided with an ultra-clean ventilation canopy opening and HVAC supply ductwork for connection to an ultra-clean ventilation canopy.

[0014] Preferably, an ultra-clean canopy is provided within the ultra-clean ventilation canopy opening and is connected to the HVAC supply ductwork.

[0015] Preferably, all services provided to the operating theatre pre-assembled module pass through a vertical face of the operating theatre pre-assembled module.

[0016] Preferably, all of the services that are provided to the operating theatre pre-assembled module pass through the receiver pre-assembled module.

[0017] Preferably, the operating theatre pre-assembled module is provided with a lamp pendant support.

[0018] Preferably, the operating theatre pre-assembled module is provided with a first pendant support, a second pendant support and a third pendant support. The provision of first, second and third pendant supports in the operating theatre pre-assembled module is advantageous because it gives the designer of an operating theatre flexibility about where to locate the pendants (typically only two pendants are needed) and it removes the need to access the operating theatre pre-assembled module after assembly to mount the pendant supports (such access not always being available).

[0019] Preferably, the receiver pre-assembled module and the operating theatre pre-assembled module each comprise an open framework.

[0020] Preferably, the service network system is for providing services to an ultra-clean operating theatre facility.

[0021] According to a second aspect of the present invention, there is provided a service network system in combination with an operating theatre facility.

[0022] According to a third aspect of the present invention, the operating facility comprises a separable operating theatre pod and a separable auxiliary rooms pod.

[0023] According to a fourth aspect of the present invention, there is provided a service network system in combination with an ultra-clean operating theatre facility.

[0024] Preferably, the ultra-clean operating facility comprises a separable operating theatre pod and a separable auxiliary rooms pod.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] Embodiments of the present invention will now be described by way of example only with

reference to the accompanying drawings where like parts are provided with corresponding reference numerals and in which:
[0026] FIG. **1** is a perspective schematic view of a service network system in combination with a traditionally built operating theatre facility;
[0027] FIG. **2** is an elevation schematic view of the service network system and operating theatre facility of FIG. **1**;
[0028] FIG. **3** is a perspective schematic view of an air handling unit (AHU) that forms part of the service network system of FIG. **1**;
[0029] FIG. **4** is a perspective schematic view of a roof pre-assembled module (PAM) that forms part of the service network system of FIG. **1**;
[0030] FIG. **5** is a perspective schematic view of a receiver pre-assembled module (PAM) that forms part of the service network system of FIG. **1**;
[0031] FIG. **6** is a perspective schematic view of a roof PAM header that forms part of the service network system of FIG. **1**; and
[0032] FIG. **7** is a perspective schematic view of an operating theatre pre-assembled module (PAM) that forms part of the service network system of FIG. **1**;
[0033] FIG. **8** is a schematic plan view of the service network system of FIG. **1**; and
[0034] FIG. **9** is a perspective schematic view of a service network system in combination with a modular operating theatre facility.
DETAILED DESCRIPTION
[0035] FIGS. **1** and **2** are schematic views of a service network system **1** according to an embodiment of the present invention for providing services to an ultra-clean operating theatre facility **3** that is located within a larger building, typically a hospital. The service network system **1** is also suitable for the provision of services to conventional operating theatre facilities, e.g. those that are not provided with ultra-clean ventilation systems. The services provided by the service network system **1** are, for example, ventilation (supply and extraction), electrical power and electrical control signals, water and medical gasses.
[0036] The operating theatre facility **3** is one built using traditional construction methods and comprises an operating theatre **5** and auxiliary rooms **7**. The auxiliary rooms **7** are typically a preparation/layup room, an anaesthetic room and a scrub/gown room.
[0037] The service network system **1** comprises an operating theatre pre-assembled module (PAM) **9**, a receiver pre-assembled module (PAM) **11**, a lid pre-assembled module (PAM) **13**, a roof pre-assembled module (PAM) **15**, a low-level header **17** and an air handling unit (AHU) pod **19**. The operating theatre PAM **9**, the receiver PAM **11** and the AHU pod **19** are all cuboidal in form s and the lid PAM **13** is generally cuboidal.
[0038] The various components of the service network system **1** and the operating theatre facility **3**, e.g. the operating theatre facility rooms, the pods, modules and headers, each have an upper face and a lower face. The upper faces face upwards in use and the lower faces face downwards in use, for example as illustrated in FIG. **1**. The components also have a front face, which is the face located to the front in FIG. **1** (i.e. facing out of the page), a back face, which is the face located to the back in FIG. **1**, a left-hand side face, which is the face located on the left-hand side in FIG. **1**, and a right-hand side face, which the face located on the right hand side in FIG. **1**.
[0039] In this embodiment of the present invention an upper part of the service network system **1** is provided above a roof **21** of a building, a lower part of the service network system **1** is provided below the roof **21** and an interface part of the service network system **1** passes through the roof **21** to connect the upper part to the lower part. The lid PAM **13** provides the interface part between the upper part and lower parts.
[0040] The upper part of the service network system **1** comprises the AHU pod **19**, the roof PAM **15** and the low-level header **17**. The lower part of the service network system **1** comprises the operating theatre PAM **9** and the receiver PAM **11**. The lid PAM **13** is located between the roof

PAM **15** and the receiver PAM **11** in order to connect the upper part of the service network system **1** to the lower part.

[0041] The operating theatre PAM **9** is located directly above the operating theatre **5** of the operating theatre facility **3**. In plan view, the operating theatre PAM **9** has the same shape as the operating theatre **5**, i.e. it is rectangular, with a length, left to right, that is greater than its width, front to back. The height of the operating theatre PAM **9** is significantly lower than the height of the operating theatre facility **3**. In this example the length of the operating theatre PAM **9** is 7.3 m, the width is 4.8 m, the height of the operating theatre PAM **9** is 0.8 m and the height of the operating theatre **5** is 4.2 m.

[0042] The operating theatre PAM **9** is adjacent to and connected with the receiver PAM **11** and the receiver PAM **11** is located directly above the auxiliary rooms **7**. In plan view, the receiver PAM **11** has the same shape as the auxiliary rooms **7**, i.e. it is rectangular, with a length, left to right, that is greater than its width, front to back. The height of the receiver PAM **11** is significantly lower than the height of the auxiliary rooms **7** serving the operating theatre **5**. In this example the length of the receiver PAM is 8.1 m, the width is 4.4 m, the height of the receiver PAM **11** is 0.9 m and the height of the auxiliary rooms **7** are 4.2 m. The operating theatre PAM **9** and the receiver PAM **11** are provided with different dimensions in order to meet each of their different requirements. In this embodiment the height of the operating theatre PAM **9** is lower than that of the receiver PAM **11** because the size of the void above the operating theatre **5** is smaller.

[0043] The lid PAM **13** is located beneath the roof PAM **15**, partially above the receiver PAM **11** and partially within the receiver PAM **11**. In plan view, the lid PAM **13** has a length and width that is less than the receiver PAM **11**.

[0044] The AHU pod **19** is located above and extends over the operating theatre PAM **9** and is located above and extends partially over the receiver PAM **11**. The AHU pod **19** is located adjacent to the lid PAM **13** and to the roof PAM **15**. The left-hand end of the AHU pod **19** and the left-hand end of the operating theatre PAM **9** are located within the same vertical plane. The AHU pod **19** has a width, front to back, that is less than the width of the operating theatre PAM **9**, such that the front and back faces of the AHU pod **19** and the operating theatre PAM **9** are located in different vertical planes, for example as shown in FIG. **8**.

[0045] The AHU pod **19** is arranged to handle the air being supplied to and being extracted from the operating theatre facility **3**. It can, for example, supply cooled and filtered air to the operating theatre **5**. The heating and cooling requirements for the AHU pod **19** are provided by heat pumps **24**. Air flow to the heat pumps **24** is provided through heat pump air inlets **23** and is rejected through heat pump air outlets **25**. An air exhaust **27** is provided for rejection from the AHU pod **19** of the air extracted from the operating theatre **5** and the auxiliary rooms **7**. The AHU pod **19** provides heating ventilation and air conditioning (HVAC) through a HVAC supply connection **29**, and through a HVAC return connection **31** to the operating theatre **5** and the auxiliary rooms **7**. A services connection **33**, is used for example for the connection of electrical power and electrical signal cables within the AHU POD **19**. The present invention facilitates the use of pre-assembled wiring looms to facilitate the provision of electrical data services and electrical power throughout the service network system **1**.

[0046] The lid PAM **13** provides a means for distributing conveying components, such as electrical cables or fluid ducting, through the roof **21** of the building, so that the conveying components can be connected between apparatus located inside the building and apparatus located outside of the building, for example on the roof **21**. The lid PAM **13** thus provides a connection through the roof **21** of the building to connect the AHU pod **19** to the receiver PAM **11**.

[0047] The lid PAM **13** comprises a lid plate **35** and a riser frame **37** that extends downwardly from the lid plate **35**. The lid plate **35** is a flat plate comprising a lower surface **39** and an upper surface **41**. A seal **43** is provided on the lower surface **39** and, in use, contacts the upper surface of the roof **21**. The upper surface **41** of the lid **35** is provided with apertures to receive conveying components.

It is provided with a HVAC supply connection **45**, a HVAC return connection **47** and a services connection **49**. HVAC supply ductwork **50** and HVAC return ductwork **52** run downwardly through the lid PAM **13** from the HVAC supply connection **45** and the HVAC return connection **47** respectively.

[0048] The receiver PAM **11** is shown, for example, in FIG. **5**. It receives conveying components from the lid PAM **13** and comprises an open framework **51** made up from a lattice of steel members **53**. The framework **51** is in the form of a cuboid that has a relatively small depth as compared to its length and width. HVAC supply ductwork **55** and HVAC return ductwork **57** is located within and supported by the framework **51**. The supply ductwork **55** is provided with a HVAC supply connection **59** that is located on the upper face of the receiver PAM **11** and three HVAC intermediate connections **61**, which are for connection to the operating theatre PAM **9**. The three HVAC intermediate connections **61** are located on the left-hand side face of the receiver PAM **11** (i.e. the face facing out of the page in FIG. **5**). The supply ductwork **55** has a plurality of outlet ducts **62** which have outlet vents which are located on the lower, left-hand, front, back, or right-hand face of the receiver PAM **11** and which, in use, provide a supply of filtered air to regions within the auxiliary rooms **7**. It is envisaged that one or more outlet vents can be provided on just one face, for example, just on the lower face, or on more than one face. The HVAC supply connection **59** is connected to the HVAC supply ductwork **50** of the lid PAM **13**. The return ductwork **57** is provided with a HVAC return connection **63** that is located on the upper face of the receiver PAM **11** and an HVAC intermediate connection **65** which is located on the left-hand side face of the receiver PAM **11**. The return ductwork **57** has a plurality of extraction ducts **66** which have extraction vents located on the lower, front, left-hand, right-hand or back face of the receiver PAM **11** and which, in use, extract air from regions within the auxiliary rooms **7**. It is envisaged that one or more extraction vents can be provided on just one face, for example, just on the lower face, or on more than one face. The HVAC return connection **63** is connected to the HVAC return ductwork **52** of the lid PAM **13**.

[0049] In addition to accommodating the HVAC supply and return ductwork **55**, **57**, the framework **51** of the receiver PAM **11** also accommodates the other services that are provided by the service network system **1**. For example, electrical cables, and gas and liquid pipework can pass into the receiver PAM **11** via the services intermediate connection **49** of the lid PAM **13**, and then can pass out of the receiver PAM **11**, either downwardly into the auxiliary rooms **7**, through its lower face, or laterally into the operating theatre PAM **9**, through its left-hand face.

[0050] The roof PAM **15**, for example as shown in FIG. **6**, comprises HVAC supply ductwork **67** that runs between the HVAC supply connection **31** on the AHU pod **19** and the HVAC supply connection **45** on the lid PAM **13**, HVAC return ductwork **69** that runs between the HVAC return connection **31** on the AHU pod **19** and the HVAC return connection **47** on the lid PAM **13** and a services conduit **71** that runs between the services connection **33** of the AHU pod **19** and the services intermediate connection **49** on the lid PAM **13**.

[0051] The operating theatre PAM **9**, for example as shown in FIG. **7**, comprises an open framework **73** made up from a lattice of steel members **75**. The framework **73** is in the form of a cuboid that has a relatively small depth as compared to its length and width. HVAC supply ductwork **77** and HVAC return ductwork **79** is located within and supported by the framework **73**.

[0052] The supply ductwork **77** is provided with three HVAC intermediate connections **81** that are located on the right-hand face of the operating theatre PAM **9** and that are for connecting to the HVAC intermediate connections **61** on the receiver PAM **11**. The supply ductwork **77** has a plurality of outlet ducts **83** which have outlet vents which are located on the lower, front, back, left-hand or right-hand face of the operating theatre PAM **9** and which, in use, provide an ultra-clean supply of air to regions within the operating theatre **5**, as will be explained in further detail below. It is envisaged that one or more outlet vents can be provided on just one face, for example, just on the lower face, or on more than one face.

[0053] The return ductwork **79** is provided with a HVAC intermediate connection **85** that is located on the right-hand face of the operating theatre PAM **9** and that is for connecting to the HVAC intermediate connection **65** on the receiver PAM **11**. The return ductwork **79** has a plurality of extraction ducts **87** which have extraction vents located on the lower, front, back, left-hand or right-hand face of the operating theatre PAM **9** and which, in use, extract air from regions within the operating theatre **5**. It is envisaged that one or more extraction vents can be provided on just one face, for example, just on the lower face, or on more than one face.

[0054] In addition to accommodating the HVAC supply and return ductwork **77**, **79**, the framework **73** of the operating theatre PAM **9** also accommodates the other services that are provided by the service network system **1** and which have already passed through the receiver PAM **11**, for example, electrical cables, and gas and liquid pipework. Those other services then pass downwardly into the operating theatre **5**, through the lower face of the operating theatre PAM **9**.

[0055] The steel members **75** of the framework **73** of the operating theatre PAM **9** are arranged to produce a square ultra-clean ventilation (UCV) canopy opening **89** at its centre, within which is cradled a UCV canopy **91**. The HVAC supply ductwork **77** of the operating theatre PAM **9** is connected to the UCV canopy **91**, such that, in use, the UCV canopy **91** can create an air curtain around an operating table that is located within the operating theatre **5**.

[0056] The framework **73** is provided with two steel diagonal cross-members **93** that run across the UCV canopy opening **89** from opposite corners and which, at the point where they cross, provide a mounting for a lamp pendant support **95** at the centre of the UCV canopy **91** for a lamp (not shown) that is located above an operating table within the operating theatre **5**.

[0057] The operating theatre PAM **9** is further provided with a first pendant support **97**, that is located towards the front face of the operating theatre PAM **9**, a second pendant support **99**, that is located towards the left-hand face of the operating theatre PAM **9** and a third pendant support **101**, that is located towards the back face of the operating theatre PAM **9**.

[0058] In use, conditioned air can be supplied to the UCV canopy **91** from the AHU pod **19** via the HVAC supply ductwork provided by the various components of the service network system **1**. Air passes out of the AHU pod **19** through its HVAC supply connection **29**, through the HVAC supply ductwork **67** of the roof PAM **15**, through the HVAC supply ductwork **50** of the lid PAM **13**, through the HVAC supply ductwork **55** of the receiver PAM **11** and then through the HVAC supply ductwork **77** of the operating theatre PAM **9**. Conditioned air can also be supplied to the auxiliary rooms **7** via the outlet ducts **62** provided in the receiver PAM **11**.

[0059] Air can be extracted from the operating theatre of the operating theatre facility **3** via the extraction ducts **87** provided in the HVAC return ductwork **79** in the operating theatre PAM **9**. The extracted air passes through that HVAC return ductwork **79**, through the HVAC return ductwork **57** of the receiver PAM **11**, through the HVAC return ductwork **52** of the lid PAM **13**, through the HVAC return ductwork of the roof PAM **15** and then into the AHU pod **19** via the HVAC return connection **31**. Air can also be extracted from the auxiliary rooms **7** of the operating theatre facility **3** via the extraction ducts **66** provided in the HVAC return ductwork **57** of the receiver PAM **11**.

[0060] In an alternative embodiment of the present invention, the service network **1** can be used in combination with a modular operating theatre facility **203** which comprises a separate operating theatre pod **205** and a separate auxiliary rooms pod **7**, as shown in FIG. **9**. The auxiliary rooms pod **7** services the operating theatre pod **5** and the auxiliary rooms pod **7** typically comprises a preparation/layup room, an anaesthetic room and a scrub/gown room. The operating theatre pod **205** and the auxiliary rooms pod **207** are independent pre-fabricated modules that are located adjacent to each other and which are connected together such that, in use, medical staff can pass between them. The operating theatre pod **205** and the auxiliary rooms pod **207** are each generally in the shape of a cuboid (but they can be generally in the shape of a cube).

[0061] In a further alternative embodiment of the present invention, the service network **1** can be used in combination with a robotic operating theatre facility (not shown). A robotic operating theatre facility will be set up differently, for example it might not have an ultra-clean ventilation canopy.

Claims

1. A service network system (**1**), for providing services to an operating theatre facility (**3**), comprising, an air handling unit (**19**), an interface pre-assembled module (**13**), a receiver pre-assembled module (**11**), an operating theatre pre-assembled module (**9**), ventilation supply ductwork (**50**,**55**,**77**) and ventilation return ductwork (**52**,**57**,**79**), wherein the ventilation supply ductwork (**50**, **55**,**77**) and ventilation return ductwork (**52**, **57**,**79**) pass from the air handling unit (**9**) then through the interface pre-assembled module (**13**), then through the receiver pre-assembled module (**11**) and then into the operating theatre pre-assembled module (**9**).
2. A service network system (**1**) as claimed in claim 1, wherein the operating theatre facility (**3**) is located within a building and wherein the air handling unit (**19**) is located outside of the building, the interface pre-assembled module (**13**) is located in an external surface of the building such that it has a first part (**35**) that is located outside of the building and a second part (**37**) that is located inside of the building, the receiver pre-assembled module (**11**) is located within the building and the operating theatre pre-assembled module (**9**) is located within the building, wherein the ventilation supply ductwork (**50**, **55**,**77**) and ventilation return ductwork (**52**, **57**,**79**) pass through the interface pre-assembled module (**13**).
3. A service network system (**1**) as claimed in claim 2, wherein the air handling unit (**19**) is located on the roof of the building and vertically above the operating theatre pre-assembled module (**9**), wherein the operating theatre pre-assembled module (**9**) and the receiver pre-assembled module (**11**) are located adjacent to each other and the ventilation supply ductwork (**55**,**77**) and the ventilation return ductwork (**57**,**79**) run laterally between the operating theatre pre-assembled module (**9**) and the receiver pre-assembled module (**11**).
4. A service network system (**1**) as claimed in claim 1, wherein the interface pre-assembled module (**13**) is a lid pre-assembled module (**13**).
5. A service network system (**1**) as claimed in claim 1, wherein the operating theatre pre-assembled module (**9**) is provided with a ultra-clean ventilation canopy opening (**89**) and HVAC supply ductwork for connection to an ultra-clean ventilation canopy (**91**).
6. A service network system (**1**) as claimed in claim 5, wherein an ultra-clean canopy (**91**) is provided within the ultra-clean ventilation canopy opening (**89**) and is connected to the HVAC supply ductwork.
7. A service network system (**1**) as claimed in claim 1, wherein all services provided to the operating theatre pre-assembled module (**9**) pass through a vertical face of the operating theatre pre-assembled module (**9**).
8. A service network system (**1**) as claimed in claim 7, wherein all of the services that are provided to the operating theatre pre-assembled module (**9**) pass through the receiver pre-assembled module (**11**).
9. A service network system (**1**) as claimed in claim 1 wherein the operating theatre pre-assembled module (**9**) is provided with a lamp pendant support (**95**).
10. A service network system (**1**) as claimed in claim 1 wherein the operating theatre pre-assembled module (**9**) is provided with a first pendant support (**97**), a second pendant support (**99**) and a third pendant support (**101**).
11. A service network system (**1**) as claimed in claim 1, wherein the receiver pre-assembled module (**11**) and the operating theatre pre-assembled module (**9**) each comprise an open framework (**51**, **73**).
12. A service network system (**1**) as claimed in claim 1, for providing services to an ultra-clean operating theatre facility (**3**,**203**).

13. A service network system (**1**) as claimed in claim 1, in combination with an operating theatre facility (**3**,**203**).

14. A service network system (**1**) as claimed in claim 13, wherein the operating facility (**203**) comprises a separable operating theatre pod (**205**) and a separable auxiliary rooms pod (**7**).

15. A service network system (**1**) as claimed in claim 1, in combination with an ultra-clean operating theatre facility (**3**,**203**).

16. A service network system (**1**) as claimed in claim 15, wherein the ultra-clean operating facility (**203**) comprises a separable operating theatre pod (**205**) and a separable auxiliary rooms pod (**207**).